

Learning What Stays: Persistence-Based Semantic Map Layering for Warehouse Robot Localization

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Abstract—Maintaining sub-10 cm localization over days and weeks is a critical requirement for warehouse robots operating in densely configured spaces. This task is challenging because the environment visible to the sensors is not all static: forklifts, movable shelving, storage boxes, and staged pallets continuously change the scene across operational shifts. Existing temporal mapping methods address this by giving more weight to features that are observed frequently or recently. However, in warehouses, permanent structures which should contribute heavily for correct localization such as pillars, beams, walls are often hidden behind stacked inventory and therefore observed rarely, while temporary pallets in open areas are observed on nearly every pass. Observation frequency is thus a poor indicator of permanence in these settings. We propose a different criterion: rather than counting how often a feature is observed, we measure how long it persists. We accumulate *survival time* for each feature, defined as the total elapsed time over which it keeps reappearing at the same location with a consistent shape. When this survival time exceeds a threshold derived from the warehouse’s operational schedule, the feature is promoted to a trusted static layer used for localization. This allows real structural features like pillars, beams, and walls seen three times over 48 hours to outrank a pallet seen forty times in one hour. We evaluated the system on 12 production robots across three warehouses in continuous 24/7 operation for 14 days (168 robot-days, approximately 860 km, 8,064 missions). Mean 2D translational RMS error was 2.0 cm. Across 13 major layout changes, a naive baseline required manual recovery in all 13, a map-aging baseline degraded to approximately 6 cm, and our system held at 2.3 cm or better without map rebuilding.

I. INTRODUCTION

Warehouse robots require sub-10 cm level localization to safely navigate tight aisles, and reach precise pick-up/drop-off locations. Standard 2D SLAM systems [1], [2] build a map from laser scans and localize against it, treating every scan return as valid map evidence irrespective of the nature of the object. Many of the surfaces visible to laser scans in warehouses are dynamic. Every hour the warehouse layout changes through pallet restacking, shelving rearrangement, random box placement in different orientations on the floor. These temporary objects with changing positions and presence corrupt the map, which causes the localization accuracy to degrade [3]. The failure is not just a matter of accuracy: a robot that loses its pose during a layout change stalls in the very aisle being restructured.

Several lines of work address this problem. Map-aging schemes fade old evidence so that stale features lose influence. Dynamic-removal methods [4], [5] reconstruct a static map by filtering out moving objects. Semantic data association [6] incorporates object labels into the estimation process. Meta-rooms [7] maintains a separate long-term model that distinguishes persistent structure from transient clutter. Khronos [8]

retains dynamic elements in the map and tracks when and how long each was observed. A recent survey [9] identifies distinguishing permanent structure from temporary clutter as the central open problem for long-term robot autonomy.

Most of these approaches rely heavily on observation frequency as a proxy for permanence. In most environments this is reasonable. But in warehouses, permanent structures such as pillars, beams, wall segments, and fixed shelving uprights are routinely hidden behind stacked inventory for days, so they accumulate few observations. Meanwhile, a pallet sitting in an open area is scanned on every pass. Any method that equates observation frequency with permanence will over-trust temporary objects and under-trust the stable anchors a robot actually needs.

We propose a persistence-based trust criterion that replaces observation frequency with *survival time*. Each time the robot re-observes a feature at the same location with a consistent geometric shape, orientation, and correspondence, the elapsed time since the last verified observation is added to a running total. When this total exceeds a threshold set from the warehouse’s operational cycle, the feature is promoted to a static layer that serves as a solid localization reference. A pillar seen three times over two days earns more trust than a pallet seen forty times in one hour. Rather than assuming what is permanent based on semantic class, our system lets each feature earn that status through demonstrated persistence. To our knowledge, no existing approach jointly addresses (1) permanence estimation independent of observation frequency, (2) robustness to prolonged occlusion, and (3) adaptation to site-specific operational rhythms.

II. TECHNICAL APPROACH

A. Platform and Localization Architecture

The robot carries two planar 2D LiDARs (270° field of view, 15–20 Hz) and a global-shutter RGB-D camera at approximately 30 Hz. A dual-EKF architecture separates local and global estimation. The odometry EKF fuses wheel encoders and visual odometry for smooth short-term tracking. The map EKF corrects the global pose by matching LiDAR scans against semantically labeled structural features. The contribution of this work is the trust management that determines which features the map EKF is permitted to use as anchors for localization.

An instance-segmentation model running on the camera stream classifies each detected object into one of three categories: fixed structure (pillars, poles/beams, wall segments, fixed shelving uprights), movable assets (pallets, boxes, movable racks), and agents (forklifts, people, other robots). These

labels are projected onto the LiDAR returns. Each labeled object is represented as a *feature*: a spatially registered cluster with a known centroid, orientation, and geometric extent. The classifier is a YOLOv8-based instance-segmentation model fine-tuned on warehouse imagery, running onboard at approximately 15 fps. Because promotion depends on geometric persistence rather than the initial semantic label, a misclassification (e.g., a pillar labeled as movable) delays promotion but does not corrupt the static layer; the feature still accumulates survival time and is eventually promoted if it persists.

B. Pipeline Overview

The overall flow from raw sensor data to trusted localization anchors proceeds as follows. The segmentation model labels each object in the camera stream, and these labels are projected onto the corresponding LiDAR points to produce semantically tagged clusters. Each new cluster is checked against existing features in the map using a geometric consistency test on its centroid, orientation, and shape. If a match is found, the feature's survival time is updated; if not, a new feature is created in the dynamic layer. Features accumulate survival time through repeated geometrically verified re-observations, and are promoted to the static layer once they exceed a site-derived persistence threshold. Only features in the static layer are used as anchors by the map EKF for global pose correction. Features that disappear from the scene are removed after two confirmed absences. The following subsections describe each stage in detail.

C. Geometric Consistency Verification

Before a re-observation is accepted, it must pass a three-part consistency check against the stored feature: (1) the centroid displacement must fall below a site-calibrated threshold, (2) the principal-axis orientation must match within tolerance, and (3) the Hausdorff distance between the observed and stored shapes must not exceed a maximum admissible value. These checks ensure that a different object placed at the same location does not inherit the trust accumulated by a previous feature or the same but displaced feature does not qualify as an anchor.

D. Two-Layer Map and Promotion

The map is organized into two layers. The *static layer* holds features that the robot treats as permanent landmarks for localization. The *dynamic layer* holds recently detected geometry whose permanence has not yet been established.

Each feature j in the dynamic layer maintains an observation count n_j and an accumulated survival time $T_{p,j}$. When the robot re-observes geometrically consistent feature j at time t , with t_j^- denoting the time of the last verified observation, both are updated:

$$n_j \leftarrow n_j + 1, \quad T_{p,j} \leftarrow T_{p,j} + (t - t_j^-) \quad (1)$$

The key property of $T_{p,j}$ is that it grows with the time span between observations, not with their count. A feature observed twice, 48 hours apart, accumulates $T_{p,j} = 48$ h. A

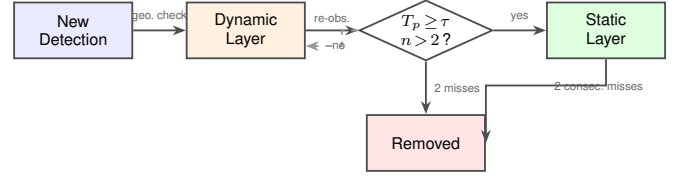


Fig. 1. Feature lifecycle in the persistence-based trust system. New detections enter the dynamic layer after passing geometric consistency verification. A feature is promoted to the static layer once its survival time T_p exceeds the site-derived threshold τ with at least two verified re-observations. Features in either layer are removed after two consecutive misses within sensor field of view. Occlusion does not count as a miss.

feature observed 50 times within one hour accumulates only $T_{p,j} = 1$ h. Promotion to the static layer requires:

$$T_{p,j} \geq \tau \wedge n_j > 2 \quad (2)$$

The survival-time threshold τ is derived from the warehouse's operational schedule: it is set to approximately how often pallets and movable assets are typically repositioned at each site (6 hours at Site A, 48 hours at Sites B and C). The $n_j > 2$ condition prevents promotion based on too few observations. Fig. 1 illustrates the full feature lifecycle.

E. Sensitivity to τ

Since τ is set from operations rather than tuned for localization, we verified robustness to its value. At $\tau/2$, localization RMS increased by approximately 0.5 cm at Site A and 0.4 cm at Site B, as some transient features were promoted too early. Site C showed no change given its already large τ . At 2τ , RMS was unchanged at all sites: doubling the threshold slowed promotion but did not affect accuracy. The system is thus not sensitive to the precise value of τ . In environments without a known operational schedule, τ could be estimated from the empirical lifetime distribution of map features during an initial observation period, promoting features that outlast a chosen percentile (e.g., the 90th) of observed lifetimes.

F. Removal and Failure Modes

A feature is removed from the static layer if the robot fails to observe it on two consecutive passes during which it was within sensor range. Occlusion does not count as a miss: if a pillar is blocked by a newly stacked pallet, it retains its trusted status. A miss is registered only when an unoccluded ray passes through the feature's expected location; if intervening geometry blocks the line of sight, the feature is neither confirmed nor penalized. This is by design, since the same occlusion pattern is why survival time is a better permanence signal than observation frequency.

One failure mode exists: a transient object observed once, then by coincidence re-observed after a long gap, can accumulate a large survival time. Two mechanisms limit this risk. First, the geometric consistency check prevents a different object at the same location from inheriting credit. Second, the two-miss removal rule catches incorrect promotions on subsequent passes. Over the 14-day evaluation, 7 of 93 new promotions (7.5%) were incorrect, and all 7

TABLE I
PER-SITE LOCALIZATION ACCURACY (2D TRANSLATIONAL RMS) AND
DRIFT REDUCTION RELATIVE TO THE NAIVE BASELINE.

Site	Area (kft ²)	Bcns. /kft ²	τ	Naive (cm)	Aging (cm)	Ours (cm)	Drift red.
A	121	~ 10	6 h	15.0	6.0	1.8	69%
B	100	~ 10	48 h	12.0	6.0	1.8	63%
C	175	~ 5	48 h	8.0	5.0	2.3	47%

were automatically removed with no measurable effect on localization.

III. EVALUATION

We evaluate whether persistence-based trust improves both localization accuracy and robustness to layout changes under continuous production conditions.

A. Deployment and Baselines

The system was evaluated on 12 production robots running 24/7 across three warehouses for 14 days: 168 robot-days, approximately 860 km, and 8,064 missions. Each mission ended at a surveyed floor marker (AprilTag or QR code) tied to a total-station survey, providing 8,064 ground-truth pose measurements. No additional infrastructure was installed. Two baselines ran on the same software stack: a *naive baseline* that treats all returns as permanent, and a *map-aging baseline* that decays older evidence (exponential decay with a half-life of 6h). We report 2D translational RMS error at marker locations and drift (error growth between consecutive markers, normalized per 1000 ft).

B. Results

Steady-state accuracy. Under normal operation, the three systems separate clearly (Table I). The naive baseline produced 8–15 cm RMS and frequently lost localization entirely in areas with high transient density, requiring manual recovery. The map-aging baseline maintained continuous localization at 5–6 cm. Our system achieved 1.8–2.3 cm (mean 2.0 cm) with 47–69% less drift than the naive baseline. Site C produced the weakest results across all systems: it is the largest, the most compartmentalized, and has roughly half the permanent-landmark density of Sites A and B (~ 5 vs. ~ 10 per 1000 sq ft). We attribute the gap primarily to landmark sparsity, though these factors are confounded in our deployment.

Layout changes. Over the 14 days we recorded 13 major layout changes (4 at Site A, 2 at Site B, 7 at Site C), defined as events where more than 50% of LiDAR returns in a single scan differed from the prior map. The naive baseline lost localization in all 13 cases, each time requiring manual recovery. The map-aging baseline remained localized but degraded to approximately 6 cm RMS. Our system held at 2.3 cm or better through all 13 events without map rebuilding, because its trusted static-layer landmarks survived the changes. Fig. 2 summarizes the per-site comparison.

Promotion statistics. Of 93 features newly promoted to the static layer during the 14-day deployment, 7 (7.5%) were later

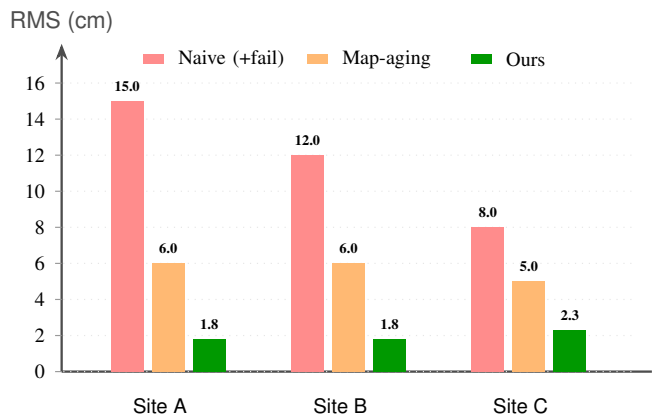


Fig. 2. Per-site localization RMS during layout-change events. The naive baseline (red) also lost localization entirely, requiring manual recovery in all 13 events. The map-aging baseline (orange) remained localized but degraded. Our system (green) held at or below 2.3 cm.

identified as incorrect. All 7 were automatically removed by the two-miss rule with no measurable effect on localization, confirming that the system self-corrects without operator intervention.

IV. DISCUSSION AND FUTURE WORK

Localization accuracy depends on having enough permanent landmarks within sensor range. In areas with few fixed structures, such as long uniform aisles at Site C, the static layer provides limited anchoring and drift increases. Improving coverage in these sparse zones, through additional fixed references or complementary sensors, is the most immediate next step.

We did not compare against offline dynamic-removal methods [4], [5] or Khronos [8], as our evaluation ran on production robots without a sensor-replay pipeline. A controlled replay dataset that enables direct comparison under identical conditions is planned as future work.

The persistence criterion relies on one domain property: warehouses have a regular cycle of change. Pallets are moved on a predictable schedule, so an object that outlasts that cycle is likely permanent. We expect this property to hold in other scheduled logistics environments such as ports, distribution centers, and hospital supply corridors. In unstructured environments without a regular change cycle, τ cannot be set from operations and would need to be learned from data. Whether a robot can discover its own τ automatically is the central open question that this work raises.

V. CONCLUSION

We presented a persistence-based trust criterion for semantic map layering in warehouse environments. The core idea is to measure permanence by how long a feature survives rather than how often it is observed. This distinction matters in warehouses, where permanent structures are frequently occluded and temporary objects are frequently visible. Twelve robots deployed across three warehouses for 14 days achieved

2.0 cm mean localization error and maintained 2.3 cm or better through all 13 major layout changes without map rebuilding. The open question for the community is whether persistence-based trust, here derived from known operational schedules, can be learned in environments where such schedules are not available.

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